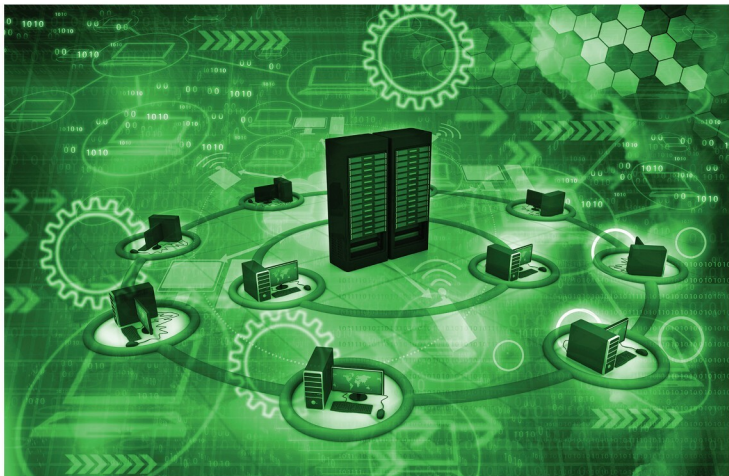


NFV: Redefining Network Infrastructure

Network Function Virtualisation or NFV aims to transform the way network operators architect their networks, by evolving standard IT virtualisation technology to consolidate many network equipment types onto industry standard high volume equipment, which could be located in data centres, network nodes and at end user premises.



Telecommunication is the backbone of today's world and it has become an integral part of our day-to-day life. As competition increases, it is important for telecom vendors to be cost effective and offer highly efficient connections with greater speed. Telecom data centres and infrastructure are governed by standards, protocols and service quality norms. Also, they are over populated by a large and increasing variety of proprietary hardware appliances. Therefore, to launch a new network service, it often requires the introduction of yet another variety of proprietary hardware, a task that is becoming increasingly difficult.

This is further compounded by the increasing costs of energy, capital, technically-trained manpower, etc. The variety of skills necessary to design, integrate and operate complex hardware-based appliances poses a real challenge to both the developer and the network operator. Moreover, hardware-based appliances rapidly reach end-of-life.

Other cost-oriented issues result in little or no revenue benefits. These constraints/limitations of hardware-based appliances (e.g., routers, firewalls, etc) has triggered the need for a new radical approach.



Figure 1: Typical network devices found in telecom data centres
(Source: <https://www.cloudsoft.io/2015/09/16/network-transformation-thinking-outside-the-middlebox>)